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August 19, 2025

Hennebery Eddy Architects, Inc.
c/o Laura Landon, AIA
109 North Rouse Avenue
Bozeman, MT 59715

RE: GEOTECHNICAL-ENGINEERING REPORT, GRTE – 4 LAZY F REHAB, MOOSE, TETON COUNTY, WYOMING
JG PROJECT NO: 25401
HENNEBERY EDDY PROJECT NO: 24041-01

Dear Laura,

We are pleased to present this Geotechnical-Engineering Report on our geotechnical site exploration for the proposed rehabilitation at historical 4 Lazy F Ranch, located north of Moose, Wyoming within the Grand Teton National Park (GTNP). This report describes site conditions observed during the subsurface exploration and presents engineering analyses and recommendations to support the design and construction of foundation elements associated with the proposed construction.

Jorgensen Geotechnical explored subsurface conditions by excavating five (5) exploratory test spanning the proposed project location. In general, the site is underlain by coarse-grained alluvium, consisting of sandy gravels and cobbles, which appear to be an adequate bearing layer for anticipated foundation loads. We recommend foundation elements be placed on the native coarse-grained alluvial soils. We observed a thin veneer of fine-grained topsoil consisting of sandy silt overlying the alluvial deposits consistently across the site. These fine-grained soils represent a risk of settlement if left beneath foundation elements. We recommend that if encountered during the excavation the topsoil or other fine-grained soils be removed from under foundation elements.

If you have any questions about this report, or if we may provide other services to you, please contact us. As the project progresses, we will be available to answer questions for you.

Respectfully submitted,

JORGENSEN GEOTECHNICAL

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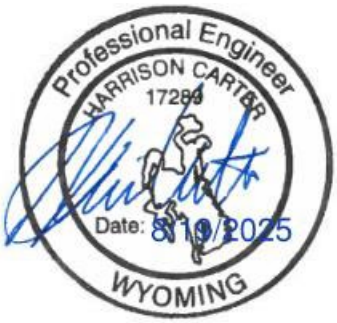
Geotechnical-Engineering Report

GRTE – 4 Lazy F Rehab| Teton County, Wyoming | Grand Teton Nation Park



43.666933°N, 110.705564°W

August 19, 2025



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1.0 REPORT SUMMARY

Jorgensen Geotechnical (JG) explored subsurface conditions by excavating five (5) exploratory test pits spanning the project site. No soil samples were obtained at the time of the exploration due to the coarse-grained conditions encountered at relatively shallow depths and our experience with similar soil conditions. Generally, the site was underlain by alluvium consisting coarse-grained sandy gravels and cobbles beneath a layer of topsoil. A detailed description of the subsurface exploration and graphical logs are shown in Appendix B.

We identified the following as specific geotechnical-engineering considerations for this project site:

- site soils: topsoil and coarse-grained alluvium
- all soils overlying the coarse-grained alluvium are assumed to be non-structural for building loads, and therefore require removal,
- groundwater was not observed. Standpipe piezometers were installed for future groundwater monitoring efforts.
- seismic and fault-related hazards exist.

The coarse-grained alluvium appears to be an adequate bearing layer for anticipated foundation loads. We recommend all topsoil and fine-grained soils (if encountered) be removed, and foundation elements be placed on the native coarse-grained alluvium.

2.0 INTRODUCTION

Jorgensen Geotechnical, LLC (JG), was commissioned by Hennebery Eddy Architects, Inc. to perform a subsurface exploration for the rehabilitation of facilities and infrastructure at the historical 4 Lazy F Ranch (4LF) in Teton County, Wyoming (Figure 1). Our specific scope of services is described in the proposal letter dated October 23, 2024. The purposes of the exploration were to provide information and geotechnical-engineering recommendations pertaining to:

- Soil Classification
- Groundwater conditions
- Site preparation and earthwork
- Seismic site classification per the International Building Code
- Evaluation of settlement potential
- Lateral Earth Pressures
- General foundation design and construction recommendations
- Slabs-on-grade design and construction recommendations

The scope of services for this project included excavating and logging five (5) exploratory test pits which ranged in depth from 5.1 to 9.2-ft below ground surface (bgs). Figures are presented in Appendix A, and a detailed description of the subsurface exploration and graphical logs are shown in Appendix B.

3.0 PROJECT DESCRIPTION

It's our understanding that the primary purpose of the project is to rehabilitate facilities and infrastructure across the historical 4 Lazy F Ranch (4LF) and to utilize the existing buildings as housing for season GTNP employees. The buildings of interest include the main lodge, eight sleeping cabins, one garage, and seven outbuildings.

Rehabilitation efforts will include roof replacement, foundation repair/replacement, exterior log repairs and refinishing, pest exclusion, historic window and door repairs, updating electrical, telecommunications, and plumbing systems to modern codes, installation of new fire suppression, and other important preservation work on all buildings.

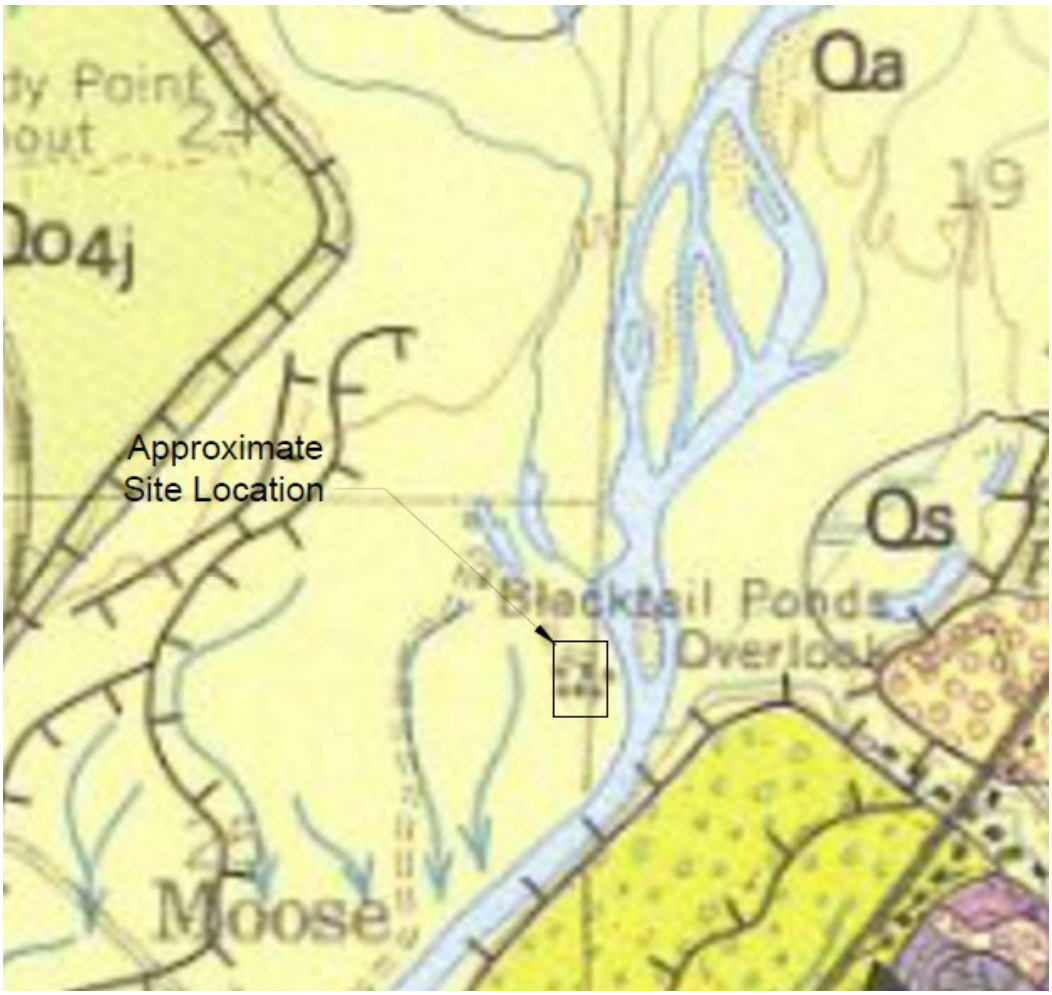


Figure 1: Site Location and Geologic Map

4.0 SITE CONDITIONS

4.1 Description

JG developed the following description of the site based on site visits, meetings with the architect, our review of available historical aerial imagery, and observations made during the subsurface exploration.

Item	Description
Location and Parcel Information	The project site is located at the end of 4 Lazy F Ranch Road in Teton county, Wyoming. It is located on parcel GRTE-00-01. National park lands surround the site. The site is bound by the Snake River to the east.
Current Ground Cover	Based on a review of historical aerial imagery, no major improvements have occurred at the site. Current ground cover consists of grasses and sage brush. To limit disturbance across the site, the excavation equipment path was determined prior to excavation. All changes to the path were approved by Nation Park Services (NPS) personnel on site.
Existing Topography	According to public topographic data on Teton County GIS the site is relatively flat at an elevation of approximately 6471-ft AMSL. We recommend getting test pit locations surveyed for site specific elevations.
Geologic Setting	▪ The project site is found on the Geologic Map of Grand Teton National Park, Teton County, Wyoming (Love et al., 1992) which is adapted as Figure 1. The map shows the location of surficial deposits and geologic structures (i.e., faults and folds). The map indicates the project site is covered by Quaternary-aged alluvium (Qa) deposits originating from the Snake River Floodplain. ▪ Bedrock was not encountered in any test pits during the subsurface investigation. The depth to bedrock is estimated to be deep. ▪ Numerous Quaternary faults (i.e., relatively young and potentially active) are mapped throughout the area, most notably, the Teton Fault system, approximately 4 miles west of the site (Zellman, 2019) with slip rates typically ranging from 0.2 to 1 mm/year. Older faults mapped nearby are believed to be old and inactive.

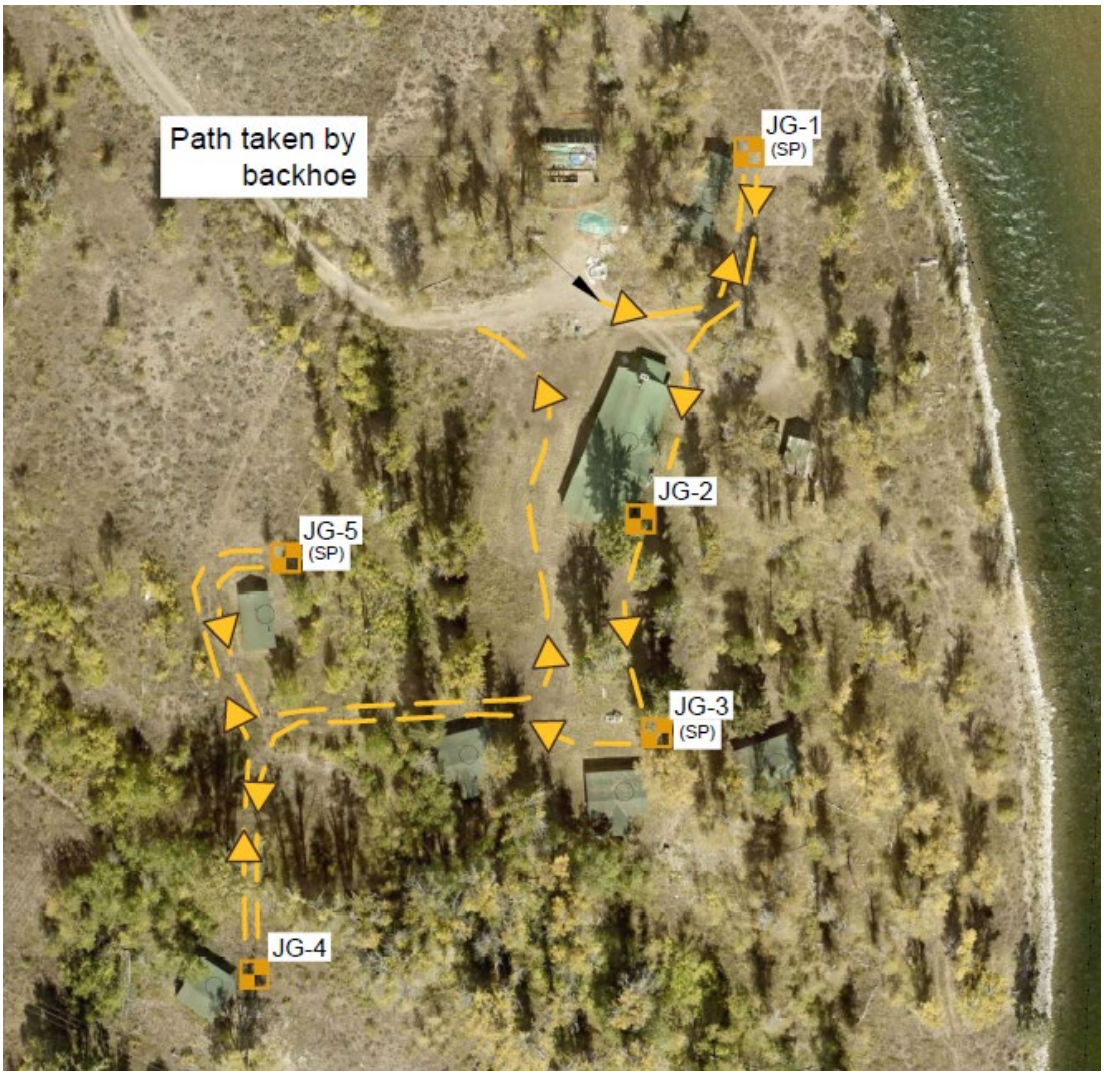


Figure 2: Test Pit Location Map

4.2 Subsurface Soil Conditions

Generally, two soil layers were observed in the test pits; topsoil and alluvium comprising sand, gravels and cobbles. The coarse-grained alluvium was observed to full depth in all test pits. Detailed test pit logs are represented graphically in Appendix B.

Layer	Layer Name	Description	Geotechnical Considerations and Uses
1	**Topsoil**	Sandy Silt • Dry • Brown to dark brown • Containing organics • Thicknesses ranging from 0.3-ft to 0.9-ft thick. • Observed in all test pits	• Potential frost heave • Potential settlement • Non-Structural applications

5	**Coarse-grained Alluvium**	<u>Poorly-Graded Sand</u> • Dry • Tan • No apparent structure • 5% gravel • 85% sand • 10% fines • Observed in JG-5 • Depths from 0.3-ft to 3.0-ft <u>Poorly-Graded Gravel with Sand and cobbles</u> • Dry to slightly moist • Light brown • Loose to very dense • Clast supported, stratified • 5-45% cobbles and boulders • 65-70% gravel • 30-35% sand • Observed in all test pits • Extended to the bottom of each test pit	• Adequate bearing layer • Structural applications

4.3 Groundwater

Groundwater was not encountered at the time of the subsurface exploration. July is generally a time of the year where groundwater levels are anticipated to be at or just past peak levels. Groundwater levels fluctuate in response to seasonal snowmelt, local irrigation, and flow levels in nearby rivers and streams.

Groundwater is anticipated to impact excavations deep excavation on this site, particularly for utility installation. Excavation related to foundation are less likely to be impacted by groundwater. The severity of the impact will depend on the excavation depth and time of year excavation is completed. If excavation occurs during the spring or early summer when groundwater levels are anticipated to be highest, the contractor should be prepared to manage groundwater during excavation, possibly through a dewatering system.

Standpipe piezometers were installed in three of the five test pits during the subsurface exploration. We recommend monitoring groundwater during Spring 2026 runoff. If you elect to monitor groundwater, we will install data loggers from Heron Instruments to continuously record groundwater levels. This monitoring will provide site-specific groundwater elevation data throughout the 2026 spring runoff period and into the summer months. Monitoring will continue until levels begin to recede and we will conclude no later than mid-autumn. A final groundwater monitoring report will be prepared upon completion of the monitoring period.

4.4 Earthquakes and Ground Shaking

The project site is located on the eastern flank of the Intermountain Seismic Belt, a zone of seismicity that extends from southern Utah through eastern Idaho, western Montana, and Western Wyoming (Smith and Arabasz, 1991). Predicted recurrence intervals for maximum credible earthquakes have

passed for most of the fault systems capable of generating magnitude 7.5 earthquakes in western Wyoming (Case, 1997), implying the risk of major earthquakes is relatively high. The owner should be aware that in the event of a large magnitude earthquake (i.e., approximately 7.5), strong ground shaking (Smith, et al., 1993) may occur.

Ground motion accelerations should be derived for the project site in accordance with the general procedure defined in the International Building Code (IBC). The IBC references ASCE 7-22 to determine the Site Class and ground motion accelerations. Based on our geologic knowledge of the region, this site is classified as **Class D – Medium dense sand or stiff clay.**

The project site is located in an area of moderate seismic activity. The current modified peak ground acceleration (PGA_M) is 0.57 (ASCE 7-22). This has been used for the analysis of seismic lateral loading on retaining walls in Section 5.4.

The provisions of the IBC are intended to provide uniform levels of performance for structures depending on their intended occupancy and use, and the risk inherent to their failure. The approach adopted in the IBC is intended to provide a uniform margin of safety against collapse at the design motion. The owner should be aware that the IBC is not intended to prevent damage or loss of function during a major earthquake; it is intended to reduce the risk of loss of life. Structural design should follow the level of risk tolerable to the owner.

4.5 Geologic and Geotechnical Hazards

4.5.1 Seismic and Fault Related Hazards

The owner should be aware that in the event of a large magnitude earthquake (i.e., approximately 7.5), strong ground shaking could potentially cause damage to structures (Smith, et al., 1993). The owner may wish to consider the option of carrying earthquake insurance in addition to homeowner's insurance. Older faults mapped nearby are believed to be old and inactive, though the Teton Fault system, approximately 4 miles west of the site is considered active. Surface rupture or displacement due to faulting is **unlikely**. Liquefaction (e.g., lateral spreading and lateral flow) typically occurs in sands or clayey sands. Liquefaction potential of the sand encountered at the time of the subsurface exploration were outside of the scope of work. Analysis of liquefaction potential would add additional subsurface exploration. We estimate that footings may be below the sand we observed near JG-3.

4.5.2 Groundwater

If excavation occurs during the spring and early summer when groundwater levels are anticipated to be high, the contractor should be prepared to manage groundwater during excavation, possibly through a dewatering system. Additionally, if construction begins at a time of year when groundwater levels are high, compaction efforts may be difficult, if not impossible, if groundwater is encountered within two feet of the bottom of the excavation. Groundwater monitoring is scheduled to occur during spring of 2026. Dewatering efforts, if employed, should reduce the water level to a minimum of 2-ft below the base of the excavation.

5.0 GEOTECHNICAL RECOMMENDATIONS

5.1 Recommendations Overview

The topsoil represents a risk of potential settlement and foundations or slabs should not be placed in direct contact with them. If topsoil is exposed in the base of foundation excavations, it should be over-excavated to expose the underlying coarse-grained alluvium and replaced with an approved structural fill to reduce the risk of future settlement. The following recommendations are for all proposed structures to be rehabilitated. Subsurface conditions were consistent across the site. If subsurface conditions differ at the time of construction at any structure, JG should be notified immediately. We anticipate foundation elements will be at or near the surface of the coarse-grained alluvium. The coarse-grained alluvium was encountered at depths ranging from 0.3 to 0.9-feet bgs is anticipated to be a suitable bearing layer for foundation elements.

5.2 Earthwork

5.2.1 Site Preparation

Prior to the placement of any fill, concrete slabs, or foundation elements, the site should be cleared and stripped of topsoil, fine-grained soils, fill, and all organic debris. No brush, roots, frozen material, or other deleterious or unsuitable materials shall be incorporated in the foundation subgrade or structural fill. All exposed subgrade surfaces should be free of mounds and depressions which could prevent uniform compaction. If unexpected fills or obstructions are encountered during site clearing or excavation, such features should be removed, and the excavation thoroughly cleaned prior to backfill placement and/or construction.

During the excavation, the equipment may disturb and loosen the surface of the coarse-grained alluvium. **All disturbed areas should be compacted with a smooth-drum vibratory roller, in vibratory mode with a minimum of three passes, prior to the placement of structural fill and footing construction.** The actual number of passes should be determined by observing whether the surface is yielding after each pass. If the surface appears to be yielding, the number of passes should be increased until a non-yielding condition is observed and approved by a representative of this office.

If the subgrade preparation occurs during Spring runoff, the soil may “pump” during compaction and a non-yielding condition may not be achievable. To avoid adverse groundwater conditions, construction shall occur before or after Spring runoff, or dewatering efforts may be required.

5.2.2 Excavation and Cut Slope Stability

OSHA regulations (29CFR1926) appear to classify the sandy alluvium at the site as Type C soil. For planning and design purposes, simple cut slopes should be no steeper than 1.5H:1V. According to OSHA, any cut slope greater than 20 feet in height requires additional analysis. These recommendations are based on observations made at the time of the site exploration. The contractor shall be responsible for adherence to OSHA and other safety regulations by observing soil conditions at the time of construction.

5.2.3 Fill Material Types and Compaction Requirements

We recommend the following uses and compaction requirements for anticipated site soils:

Table 5-1: Soil Uses and Compaction Requirements

Location/Use	Soil type	Minimum Compaction Requirement	Moisture
Structural[1,2]	Stony alluvium or imported pit-run	Proof roll to observe Non-yielding condition	Usually 6-8%[2]
	Imported crush or Sandy alluvium	95% ASTM D698	Usually 6-8%[2]
Non-Structural[3]	Topsoil	92% ASTM D698	+2-2% OMC

1. Structural uses include but are not limited to foundation and slab subgrade
2. All structural soils may be used for non structural uses, as well
3. Non-structural uses include but are not limited to utility trench backfill, exterior backfill, and general site grading
4. Clean, stony soils are typically moisture insensitive and field observations are usually adequate to estimate when soils achieve the proper moisture.

We recommend compaction of structural subgrade soils be observed and tested during construction. If stony fill is observed to be in a yielding condition or non-stony fills do not achieve the recommended compaction requirements, the area represented by the test should be reworked and retested as required to meet the requirements.

Compaction testing of site-derived coarse-grained soils containing more than approximately 30% retained on the ¾-in sieve, or imported stony soils, (e.g. “pit-run,”) with a nuclear density gauge is usually problematic due to the presence of large stones. Therefore, we recommend compacting stony fills using a **method specification**, for which Table 5-2 provide initial guidelines.

Table 5-2: Compaction Parameters for Native Coarse-grained Alluvium

Compactor Type	Lift Thickness	Maximum Particle Size	Minimum Number of Passes[1]
5-ton vibratory	12 inches	9-inch[2]	3
1.5-ton vibratory	9 inches	6-inch	5
Hand-held	4 inches	4-inch	5

1. The actual number of passes should be determined by observing whether the surface is yielding after each pass. If the surface appears to be yielding, the number of passes should be increased until a non-yielding condition is observed.
2. Occasional clasts to 12-inch are permitted, if encountered, but should not be nested.

The method specification may be established as follows:

- The contractor will place fill in loose lifts no greater than specified in the above table for whichever class of compactor is used.
- Fill will be compacted with the *minimum* number of passes specified above. The actual number of passes should be determined by observing compaction after each pass to determine if the surface is non-yielding. If the fill surface appears to be yielding, the number of passes should be increased until a non-yielding condition is observed.
- Once the number of passes is determined, this **method** (unique to the material type, compactor, lift thickness, and number of passes) may be continued for the rest of the project as long as fill material properties and subgrade soil conditions remain the same.

It is important to establish a method specification as early in the construction as possible and apply it consistently for the entirety of the project. JG should observe lift thickness, number of passes, and equipment used during compaction. Additional guidance on construction observations may be found in Section 5.2.7.

5.2.4 Final Backfilling and Grading

Properly compacted backfill and site drainage should be considered. Structural and non-structural fill material may be used for backfilling and grading. Compaction guidelines for both soil types can be found in Section 5.2.3. Final grading should provide positive drainage of at least 0.5 foot in the first 10 feet away from all structures. Adequate gutters are strongly recommended. Roof runoff should be discharged at least 3 feet away from the building or exterior slabs. Swales or other moisture collection points should be avoided within 20 feet of the footings. Drainage swales should slope a minimum of 2%.

Exterior fills should be placed as early as possible to reduce moisture infiltration along foundation walls. However, do not over-compact exterior backfills against “green” foundation walls. Utility trenches should also be backfilled in lifts and compacted with the same care as exterior backfills, lest the fill settles causing damage to overlying landscaping, hardscapes, etc.

5.2.5 Crawlspace Ventilation and Radon

Evaluation of radon was beyond the scope of work; local codes should be followed and specialty contractors employed, if necessary. The building contractor is ultimately responsible for following local building codes. Ventilation to reduce moisture and potential accumulation of radon gas is required by code for inhabited and uninhabited spaces below grade. A capillary break layer, as described in Section 5.5.1, may also accommodate a radon vent pipe.

5.2.6 Reinforcing, Utilities Testing, and Concrete Considerations

Footings, slabs, and foundation walls should be reinforced to resist differential movement. Consultation with a Structural Engineer to specify adequate reinforcement is suggested. Water and sewer lines should be pressure tested before backfilling. Exterior concrete should contain 5% to 7% entrained air. ***Note***: Minor cracks in the foundation walls, floor slabs, and sheetrock are normal and should not be a cause for concern.

5.2.7 Observation during Construction

Recommendations in this report are contingent upon our involvement. If any unexpected soils or conditions are revealed during construction, this office should be notified immediately to survey the conditions and make necessary modifications. All excavations and foundation subgrades should be

observed by a representative of JG prior to fill or concrete placement, especially if questionable materials are exposed. Notice shall be provided at a minimum of 24 hours before the requested observation.

We can provide the most value observing site conditions at the following times:

1. Upon completion of site preparation to verify all organics and unsuitable material have been removed, and to verify alluvial soils have been reached and structural fills have been prepared in accordance with Section 5.2.1.
2. We can observe placement and compaction of fill (if used) and aid in the development of a method specification, Section 5.2.3.
3. We can perform density testing by means of proof rolls and with the use of a Nuclear Density Gauge for structural and non-structural soils.

5.3 Shallow Foundations

We recommend the following considerations for shallow foundation systems, assuming recommendations for site preparation, over-excavation, and compaction are followed:

Item	Description	
	Topsoil	**Coarse-grained Alluvium/ Imported Structural Fill**
Bearing Layer	Not recommended as a bearing layer	Coarse-grained alluvium, or approved structural fill placed in contact with native coarse-grained alluvium
Settlement	Subject to potential settlement	Significant settlement (i.e., greater than 1-inch of total settlement or greater than 0.5-inches of differential settlement) is not anticipated, assuming the site is prepared according to the recommendations of this report.
Bearing Capacity[1,2]	Not recommended as a bearing layer	4,000 psf
Frost Depth	Footings should be placed at a minimum depth of 34 inches below finished grade, with a minimum foundation exposure of 6 inches above finished grade.	
Ultimate Soil Friction	tan(30°) = 0.58 – for the interface of cast-in-place concrete on structural fill	

1. Soil parameters (i.e., inputs to the bearing capacity equation) were derived based on local knowledge and experience working with similar soil deposits. Bearing capacity of sand is the limiting bearing capacity and is presented above.
2. Soil bearing capacity is dependent not only on the soil strength, but also the geometry of the foundation elements. The calculated bearing capacity assumes 2-ft wide strip footings placed a minimum of 3-ft feet bgs, all fine-grained soil has been removed from beneath foundation elements, and coarse-grained alluvium or imported “pit-run” material has been properly compacted. We estimated groundwater at a depth of 3-ft below ground surface. If footing size and depth differs from these assumptions, this office should be notified to evaluate the foundation configuration. It is often the case that heavily loaded, isolated footings may be optimized (i.e., made smaller) using a larger

bearing capacity, thereby reducing the quantity of concrete required. Please contact JG for an evaluation.

5.4 Lateral Earth Pressures

Lateral pressures were calculated using methods suggested by Bowles (1996) for at-rest, active, and passive conditions and are presented in Table 5-3. These values assume native coarse-grained alluvium will be used as non-structural backfill and placed and compacted in a moist state. We have assumed an estimated internal friction angle of 35° and a unit weight of 135 pcf based on experience working with similar soil deposits. Calculations assume level backfill against foundation walls or retaining walls.

Table 5-3: Lateral Pressure Parameters for Native Coarse-grained Sandy Alluvium

Condition	Coefficient of Earth Pressures	γK (equivalent fluid pressure)
Static Conditions Level Backfill	K_o = 0.43 K_a = 0.27 K_p = 3.69	58 pcf 37 pcf 498 pcf
Earthquake Conditions Level Backfill	K_{ae} = 0.47 K_{pe} = 3.08	64 pcf 416 pcf

Lateral earth pressure design will be estimated based on the concept of equivalent fluid pressures, in which the soil pressure-distribution is triangular against the foundation or retaining wall. The following sections summarize each design situation.

5.4.1 Active Pressures

Application	Retaining walls, which are allowed to deflect and develop an active soil wedge
Resultant Force Calculation	½ $\gamma K_a H^2$; pounds per horizontal foot of wall
Resultant Force Location	one-third the wall height (1/3 H) above the base
Seismic Acceleration[1]	k_h = 0.29g (½ PGA_M) per the ASCE 7-22
Seismic Calculation Basis	Mononobe-Okabe equations (Bowles, 1996)
Seismic Resultant Force	½ (γK_{ae} - γK_a) H^2; pounds per horizontal foot of wall
Seismic Resultant Force Location[2]	Applied at 60% of the wall height above the base

1. Because the maximum acceleration occurs only briefly during an earthquake, it is common practice when designing dams and other earth structures to reduce the design acceleration to ½ of the maximum design acceleration (Hynes-Griffin and Franklin, 1984).
2. Research has indicated that lateral pressures due to earthquakes are non-hydrostatic in distribution, and the resultant acts above the lower third-point of the wall (Bakeer, et al, 1990).

5.4.2 At-Rest Pressures

Application[1]	Basement walls, or other walls which are restrained and not allowed to deflect

Resultant Force Calculation	½ $\gamma K_0 H^2$; pounds per horizontal foot of wall
Resultant Force Location	One-third the wall height (1/3 H) above the base

1. Design control of such walls shall be whichever generates the higher resultant force: at-rest pressures or active seismic pressures.

5.4.3 Passive Pressures

Application[1]	Toe of retaining walls where the wall is allowed to move
Resultant Force Calculation	½ $\gamma K_p H^2$; pounds per horizontal foot of wall
Resultant Force Location	One-third the wall height (1/3 H) above the base
Design Control	k_h = 0.29g (½ PGA_M) per the ASCE 7-22
Seismic Calculation Basis	Mononobe-Okabe equations (Bowles, 1996)
Seismic Resultant Force Location	One-third the wall height (1/3 H) above the base

1. Passive pressure design should neglect loose fill and soil located within the frost zone.

5.5 Slabs-on-Grade

5.5.1 Interior Slabs-on-Grade

Interior slabs should be at least 4 inches thick, and any slabs bearing vehicles should be at least 6 inches thick, or as approved by a Structural Engineer. Minor floor cracking of slab-on-grade construction is difficult, if not impossible, to prevent. Such cracking is normal and should be expected to occur with time. Buildings are almost never free of cracks, and cracking is caused by many factors other than soil movement, such as concrete shrinkage or curling, or daily and seasonal variability in temperature and humidity.

An impermeable layer (usually plastic) is suggested beneath interior slabs, underlain by 4 inches of clean drain gravel that will act as a capillary break to reduce dampness. Two options are available to reduce the tendency for the concrete to crack or curl as it dries:

1. A blotter layer may be placed under the slab. In the past, loose sand has been used for this purpose, but is no longer recommended. A cover of 4 inches of trimmable, compactible, granular material may be placed over the impermeable layer to receive the concrete slab. This material usually consists of “crusher run material”, which varies in size from about 1.5-inch down to rock dust. Alternatively, 3 inches of compacted, fine-graded material such as crusher fines or manufactured sand may be used.

2. The blotter layer may be eliminated if the concrete is reinforced properly. The attached article entitled “Controlling Curling and Cracking in Floors to Receive Coverings” provides a discussion of proper floor slab reinforcement. If the contractor needs additional guidance on reinforcement, a Structural Engineer should provide it.

Three articles from the American Concrete Institute (ACI) that discuss these options can be found in the references section (ACI, 1997; Suprenant and Malisch, 1998 & 1999). We are able to offer additional guidance if requested.

5.5.2 Exterior Slabs-on-Grade

Exterior slabs (e.g., sidewalks, patios, driveways, etc.) typically sustain the greatest damage. Cracking is almost impossible to avoid, and freeze-thaw adds to the difficulty caused by soil movement. The fine-grained soils may cause particularly severe frost damage and should be complete removed below slabs.

Exterior slabs should be at least 4 inches thick, 6 inches if supporting vehicles, or as directed by the Structural Engineer. Exterior slabs should not be tied to foundation walls. Any movement of exterior slabs may be transmitted to the foundation walls, resulting in damage. Posts for patios or other exterior columns should not bear on exterior slabs. If the slabs settle or rise, the movement can be transmitted to the post, resulting in damage to the structure.

Performance of exterior slabs in areas underlain by the topsoil soils may be improved by the complete removal of these soils and replacing them with native sandy alluvium or imported fill and placing a minimum of 6-inches of road mix gravel (e.g., WYDOT Grading H) immediately below the slab.

6.0 LIMITATIONS

This report has been prepared based on a limited amount of data. Actual site conditions may vary. These services have been performed in a manner consistent with the level of care and skill ordinarily exercised by members of the profession currently practicing under similar conditions. No other warranty is made or implied.

This report is site-specific and has been prepared in support of the proposed project. This report is for the sole use of the current property owner and their design and construction team and shall be considered non-transferable to future property owners without the written consent of Jorgensen Geotechnical. Under no circumstances are the figures and text to be used separately.

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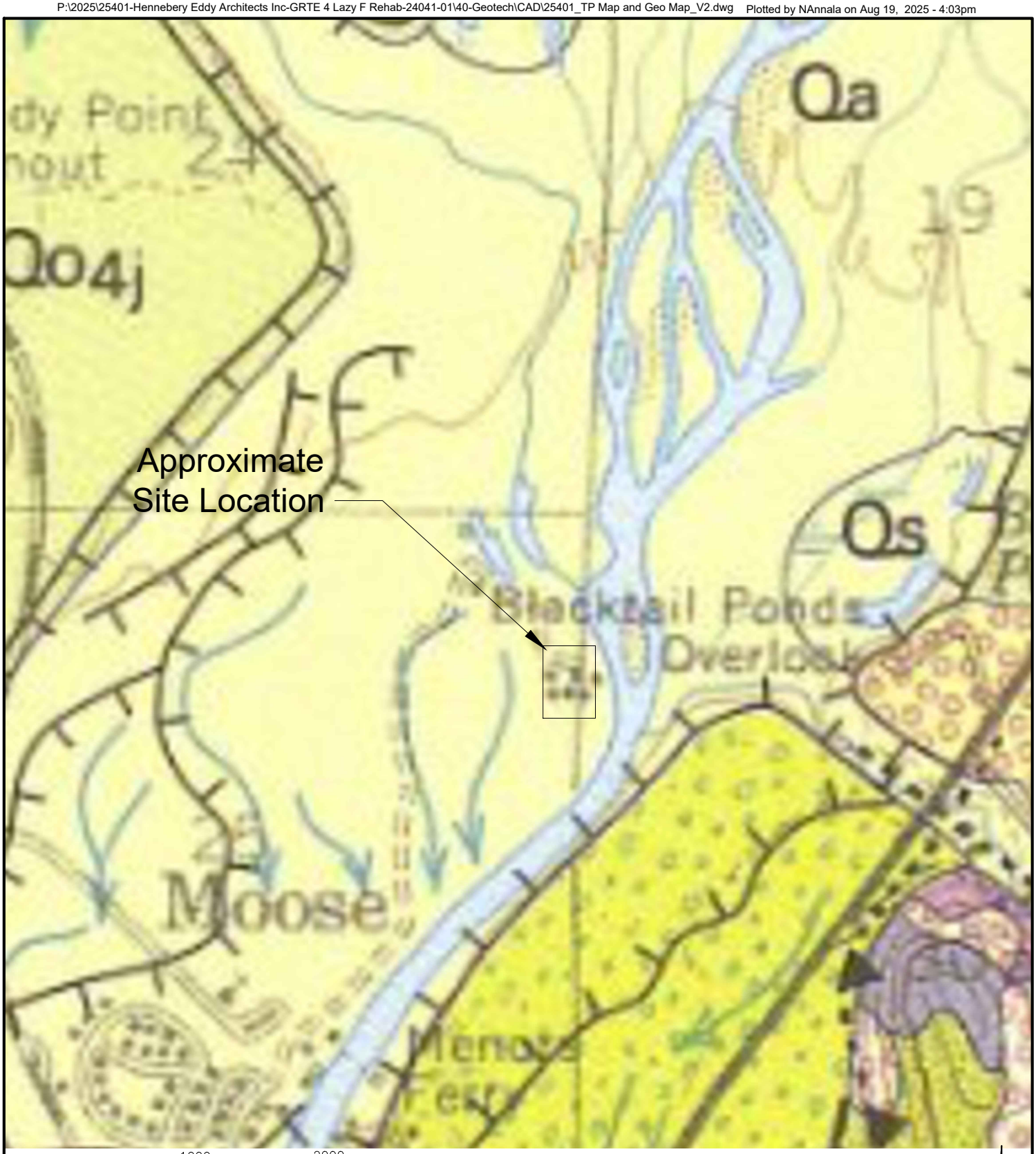
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APPENDIX A

Figures



0 1000 2000

SCALE: 1 INCH = 1000 FEET

THIS SCALE VALID ONLY FOR 8.5x11 PRINTS

Map symbols: Qa - Alluvium - stream and river deposits

From Love et. al, Geologic Map of Grand Teton National Park, 1992

DRAFTED BY: NJA REVIEWED BY: HC PROJECT NUMBER 25401	SHEET TITLE: Figure 1 Site Location and Geologic Map	PROJECT TITLE: Geotechnical-Engineering Report GRTE - 4 Lazy F Rehab Teton County, Wyoming	JORGENSEN GEOTECHNICAL, LLC PO Box 9550 | 1315 HWY 89 S., Ste. 201 Jackson, WY 83002 PH: 307.733.5150 www.jorgeng.com



0 100 200

SCALE: 1 INCH = 100 FEET
THIS SCALE VALID ONLY FOR 8.5x11 PRINTS

Test pit locations are approximate and may be surveyed upon request.

DRAFTED BY:	NJA
REVIEWED BY:	HC
PROJECT NUMBER	25401

SHEET TITLE:
Figure 2
Test Pit Location Map

PROJECT TITLE:
Geotechnical-Engineering Report
GRTE - 4 Lazy F Rehab
Teton County, Wyoming

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APPENDIX B

Subsurface Exploration Procedures and Test Pit Logs

Field Exploration: Excavated Test Pits

Test Pit Locations and Elevations

Test pit locations were determined by Jorgensen Geotechnical (JG) staff to span the proposed project site. To limit disturbance, test pits were not excavated near the most northern existing structure. Locations of the test pits are presented on Figure 2. Test pit locations JG-1 through JG-3 differed from proposed test pit locations due to existing infrastructure (i.e., utilities) and were adjusted during exploration effort. JG-1 was shifted to the east, JG-2 to the south, and JG-3 to the west. Representatives of the National Park Service approved re-location of the test pits prior to disturbance. The approximate location of each test pit was estimated based relative proximity landmarks identifiable on aerial photography. Approximate elevations were estimated based on publicly available topographic data from Teton County GIS. We recommend surveying the test pits if more precise and/or accurate coordinates or elevations are desired.

Subsurface Exploration Procedure

JG conducted a site exploration on July 16, 2025. Five (5) test pits were excavated to depths varying from 5.1-ft to 9.2-ft bgs. Test pit locations were chosen to bracket the site, and to profile soil conditions near proposed rehabilitation and new structures. Due to vegetation surrounding the site, available exploration locations were limited to reduce disturbance across the site. During excavation, the different layers of strata were separated. Soil type, consistency, and relative moisture content were observed and documented by JG personnel. Though samples were collected during the exploration, lab testing was not performed on soils encountered in the test pits. Similar soil conditions in Jackson Hole have been tested extensively by this office, and the soil's engineering properties are reasonably well-understood. Soil-engineering properties were based on visual classification and our experience with similar deposits in the area. Standpipe piezometers consisting of 4-in perforated PVC were installed into test pits JG-1, JG-3, and JG-5. Test pits were backfilled in the same layered strata they were excavated immediately after they were logged and sampled.



Photo 1: Typical Test Pit Excavation

Test Pit Logs

Field data were collected in the field digitally using the application pLog on a tablet. Draft logs were reviewed in the office, and final versions are presented below. The final logs represent JG's interpretation of the soil conditions, and, although site conditions appear consistent across the property, actual soil conditions may differ from those observed in the test pits.



Photo 2: Coarse-Grained Stony Alluvium from Test Pit JG-1



Photo 2: Coarse-Grained Sandy Alluvium from Test Pit JG-5

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Test Pit ID: JG-1

Project Name: GTRE 4 Lazy F
Project Location: Teton County, Wyoming
Client: Hennebury Eddy Architects Inc
Logged By: Nick Annala **Checked By:** HC
Project Number: 25401
Latitude: 43.667161 **Longitude:** -110.705182 **Elevation:** 6471.00
Date Started: Jul 16 2025 **Completed:** Jul 16 2025
Notes:
Contractor: FCE
Equipment: JD 310SJ
Stopped due to cave-ins.. Installed groundwater standpipe piezometer to a depth of 9.0-ft bgs with 1.0 ft of stick-up.

Depth	Graphic	Sampler Type	Number	Pocket Pen (tsf)	Material Description	Moisture Content (%)	Dry Unit Wt. (pcf)	Atterberg Limits: Liquid Limit	Atterberg Limits: Plastic Limit	Atterberg Limits: Plasticity Index	Well
0					0.00-0.90 Sandy SILT (ML): dry, brown, soft, containing organic material, 0% gravel, 45% sand, 55% fines, Varying in thickness throughout the test pit from near surface to .9 ft [TOPSOIL]						
1					0.90-9.00 Poorly Graded GRAVEL with Sand (GP): slightly moist to moist, light brown, dense to very dense, clast supported, stratified, 45% cobbles by volume, remaining 70% subrounded gravel, 30% sand, 0% fines by mass, roots extending to 2.6-ft bgs. Increased moisture with depth. [ALLUVIUM]						
2											
3											
4											
5											
6											
7											
8											
9											

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Test Pit ID: JG-2

Project Name: GTRE 4 Lazy F
Project Location: Teton County, Wyoming
Client: Hennebury Eddy Architects Inc
Logged By: Nick Annala **Checked By:** HC
Project Number: 25401
Latitude: 43.666607 **Longitude:** -110.705400 **Elevation:** 6471.00
Date Started: Jul 16 2025 **Completed:** Jul 16 2025
Notes:
Contractor: FCE
Stopped due to cave-ins. Backfilled with spoils.
Equipment: JD 310SJ

Depth	Graphic	Sampler Type	Number	Pocket Pen (tsf)	Material Description	Moisture Content (%)	Dry Unit Wt. (pcf)	Atterberg Limits: Liquid Limit	Atterberg Limits: Plastic Limit	Atterberg Limits: Plasticity Index	Well
1					0.00-0.30 SANDY SILT (ML): dry, dark brown, soft, containing organic material, 5% gravel, 40% sand, 55% fines, [TOPSOIL]						
2					0.30-6.00 Poorly Graded GRAVEL with Sand (GP): slightly moist to moist, light brown, loose, clast supported, 5% cobbles by volume, remaining 65% subrounded gravel, 35% sand, 0% fines by mass, roots extending to 3.0-ft bgs. [ALLUVIUM]						
3											
4											
5											
6											
7											
8											
9											

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Test Pit ID: JG-3

Project Name: GTRE 4 Lazy F
Project Location: Teton County, Wyoming
Client: Hennebury Eddy Architects Inc
Logged By: Nick Annala **Checked By:** HC
Project Number: 25401
Latitude: 43.666285 **Longitude:** -110.705364 **Elevation:** 6471.00
Date Started: Jul 16 2025 **Completed:** Jul 16 2025
Notes:
Contractor: FCE
Equipment: JD 310SJ
Stopped due to cave-ins. Backfilled with spoils. Installed groundwater standpipe piezometer to a depth of 8.0-ft bgs with a stick-up of 2.0-ft.

Depth	Graphic	Sampler Type	Number	Pocket Pen (tsf)	Material Description	Moisture Content (%)	Dry Unit Wt. (pcf)	Atterberg Limits: Liquid Limit	Atterberg Limits: Plastic Limit	Atterberg Limits: Plasticity Index	Well
0					0.00-0.40 SANDY SILT (ML): dry, dark brown, soft, containing organic material, 5% gravel, 40% sand, 55% fines, [TOPSOIL]						
0.4					0.40-8.00 Poorly Graded GRAVEL with Sand (GP): dry to moist, light brown, loose to medium dense, clast supported, 15% cobbles by volume, remaining 65% subrounded gravel, 35% sand, 0% fines by mass, [ALLUVIUM]						
8											

1
2
3
4
5
6
7
8
9

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Test Pit ID: JG-4

Project Name: GTRE 4 Lazy F
Project Location: Teton County, Wyoming
Client: Hennebury Eddy Architects Inc
Logged By: Nick Annala **Checked By:** HC
Project Number: 25401
Latitude: 43.665916 **Longitude:** -110.706194 **Elevation:** 6471.00
Date Started: Jul 16 2025 **Completed:** Jul 16 2025
Notes:
Contractor: FCE
Stopped due to cave-ins. Backfilled with spoils.
Equipment: JD 310SJ

Depth	Graphic	Sampler Type	Number	Pocket Pen (tsf)	Material Description	Moisture Content (%)	Dry Unit Wt. (pcf)	Atterberg Limits: Liquid Limit	Atterberg Limits: Plastic Limit	Atterberg Limits: Plasticity Index	Well
0					0.00-0.70 Sandy SILT (ML): dry, brown, soft, containing organic material, 0% gravel, 40% sand, 60% fines, Varying in thickness from near surface to .7 ft [TOPSOIL]						
1					0.70-5.10 Poorly Graded GRAVEL with Sand (GP): dry to moist, light brown, medium dense, clast supported, 20% boulders, 35% cobbles by volume, remaining 65% gravel, 35% sand, 0% fines by mass, subrounded, Test pit caving in at 3-ft to 4 ft [ALLUVIUM]						
2											
3											
4											
5											
6											
7											
8											
9											

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1315 HWY 89 S., Suite 201 - Jackson, WY 83002
Telephone: 307.733.5150

Test Pit ID: JG-5

Project Name: GTRE 4 Lazy F
Project Location: Teton County, Wyoming
Client: Hennebury Eddy Architects Inc
Logged By: Nick Annala **Checked By:** HC
Project Number: 25401
Latitude: 43.666545 **Longitude:** -110.706136 **Elevation:** 6471.00
Date Started: Jul 16 2025 **Completed:**
Contractor: FCE
Equipment: JD 310SJ

Notes:
Hit septic line at approximately 2.6 ft below grade, extending through the test pit 2-ft to the north. Standpipe piezometer installed to a depth of approximately 9'-2" bgs.

Depth	Graphic	Sampler Type	Number	Pocket Pen (tsf)	Material Description	Moisture Content (%)	Dry Unit Wt. (pcf)	Atterberg Limits: Liquid Limit	Atterberg Limits: Plastic Limit	Atterberg Limits: Plasticity Index	Well
0					0.00-0.30 SANDY SILT (ML): dry, brown, soft, no apparent structure, 0% gravel, 40% sand, 60% fines, [TOPSOIL]						
1					0.30-3.00 Poorly Graded SAND (SP): dry, tan, loose, no apparent structure, 5% gravel, 85% sand, 10% fines, [ALLUVIUM]						
2											
3					3.00-9.20 Poorly Graded GRAVEL with Sand (GP): dry to slightly moist, light brown, medium dense to dense, clast supported, 15% cobbles by volume, remaining 65% gravel, 35% sand, 0% fines by mass, subrounded, [ALLUVIUM]						
4											
5											
6											
7											
8											
9											